uterine wall through the umbilical cord. It is the organ by which the nutritive, respiratory and excretory functions are fulfilled. The embryo's heart develops during the fourth week of pregnancy and circulates blood through the umbilical cord and placenta as well as through its own tissues.

The primary germ layers serve as the primitive tissues from which all body organs develop. The ectoderm gives rise to the central nervous system (brain and spinal cord), peripheral nervous system, epidermis and its derivatives and mammary glands. The connective tissue, cartilage and bone, muscles, organs of urinogenital system (kidney, ureter and gonads) arise from the mesoderm. The endodermal derivatives are **e**pithelium of gastrointestinal and respiratory tract, liver, pancreas, thyroid and parathyroids.

Human pregnancy lasts for about 280 days or 40 weeks and is called the **gestation period**. It can be divided for convenience into three trimesters of three months each. The **first trimester** is the main period of organogenesis, the body organs namely the heart, limbs, lungs, liver and external genital organs are well developed. By the end of the **second trimester**, the face is well formed with features, eyelids and eyelashes, eyes blink, body is covered with fine hair, muscle tissue develops and bones become harder. The foetus is fully developed and is ready for delivery by the end of nine months (**third trimester**).

During pregnancy, the placenta acts as a temporary endocrine gland and produces large quantities of **human Chorionic Gonadotropin (hCG), human Chorionic Somatomammotropin (hCS) or human Placental Lactogen (hPL), oestrogens and progesterone** which are essential for a normal pregnancy. A hormone called **relaxin** is also secreted during the later phase of pregnancy which helps in relaxation of the pelvic ligaments at the time of parturition. It should be noted that hCG, hPL and relaxin are produced only during pregnancy. In addition, during pregnancy the level of other hormones like oestrogen and progesterone, cortisol, prolactin, thyroxine, etc., is increased several folds in the maternal blood. These hormones are essential for supporting foetal growth.



A female uterus is normally about 3 inches long and 2 inches wide but can expand 20 times during pregnancy.

2.7 Parturition and lactation

Parturition is the completion of pregnancy and giving birth to the baby. The series of events that expels the infant from the uterus is collectively called "labour". Throughout pregnancy the uterus undergoes periodic episodes of weak and strong contractions. These contractions called **Braxter-Hick's contractions** lead to false labour. As the pregnancy progresses, increase in the oestrogen concentration promotes uterine contractions. These uterine contractions facilitate moulding of the foetus and downward movement of the foetus. The descent of the foetus causes dilation of cervix of the uterus and vaginal canal resulting in a **neurohumoral reflex** called **Foetal ejection reflex** or **Ferguson reflex**. This initiates the secretion of oxytocin from the neurohypophysis which in turn brings about the powerful contraction of the uterine muscles and leads to the expulsion of the baby through the birth canal. This sequence of events is called as **parturition** or **childbirth**.

Relaxin is a hormone secreted by the placenta and also found in the corpus luteum. It promotes parturition by relaxing the pelvic joints and by dilatation of the cervix with continued powerful contractions. The amnion ruptures and the amniotic fluid flows out

TN_GOVT_XII_Zoology_chapter 2.indd 24 24-02-2020 4.00.22 PM